Adaptive Digital Wellbeing Coach

Data Preparation and Model Exploration

Samsung Innovation Campus | Group 8

Data preparation: İsmail Hakkı Sarıoğlan
Model development: Hayrettin Kaan Özsoy

The project investigates prediction of elevated smartphone use in the next 30-minute window, with later personalization and user feedback. This report presents the completed data preparation work and the current offline comparison of a fixed rule, Logistic Regression and Random Forest. Random Forest is the main candidate for further research, its performance advantage is modest and does not establish deployment readiness or reduction of unwanted use.

Report overview

The report is organized from data preparation and feature engineering to preliminary model exploration. Part 1 presents the cleaning rules, exploratory findings, candidate features and quality checks. Part 2 presents model selection, training, evaluation and implementation, with the experimental settings specified alongside the results.

The prediction task uses completed-window behaviour to forecast the immediately following 30-minute window. Operational proxy labels are distinguished from self-reported unwanted use. The current phase establishes data preparation and offline model exploration, personalization and Android deployment remain subsequent stages.

Part 1 | Data Preparation and Feature Engineering

1. Overview

The data pipeline converts irregular, multi-sensor event logs into dataset v1, a 30-minute window table. It addresses clock errors, duplicate unlocks, missed screen-off events and feature leakage. Its reported outputs are 366,747 valid sessions, 443,781 total windows, 372,349 valid windows and 366,362 target-bearing windows.

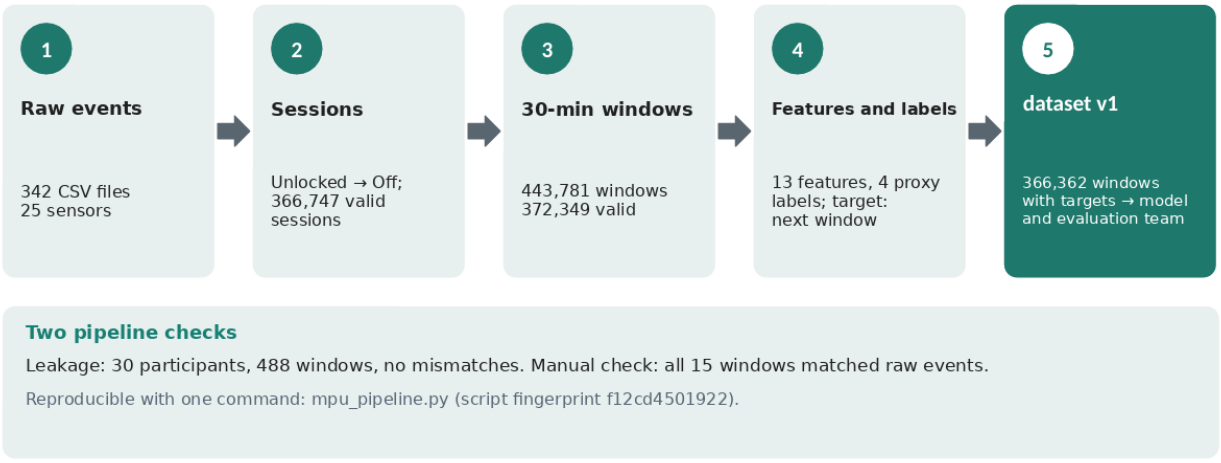


Figure 1. Dataset v1 processing stages and validation summary. Original data preparation figure.

2. Data Collection

The source is the CRAWDAD telefonica/mobilephoneuse dataset, attributed to Pielot, Cardoso, Katevas, Serrà, Matic and Oliver (2017), DOI 10.15783/jh3s-h006. The data report describes 342 volunteer Android users in Spain, recorded during April–July 2016. Ages range from 18 to 66, with mean age 37.9; median valid recording duration is 22.7 days per participant.

The 25 sensor streams include screen and app events, notifications, proximity, orientation, activity, battery, network, environmental and ESM records. The pipeline loads only ts_raw, sensor_id, Screen_Value, App_Value and Notif_Action. Millisecond timestamps are converted to local epoch seconds using participant timezone metadata plus a one-hour summer-time adjustment, then stably sorted.

The source study sent 10–15 ESM notifications per day, which may have influenced natural behaviour. The project keeps raw CSVs out of redistribution and shares derived outputs; any wider redistribution remains subject to the dataset terms. Historical Android records are not assumed to represent current Turkish users.

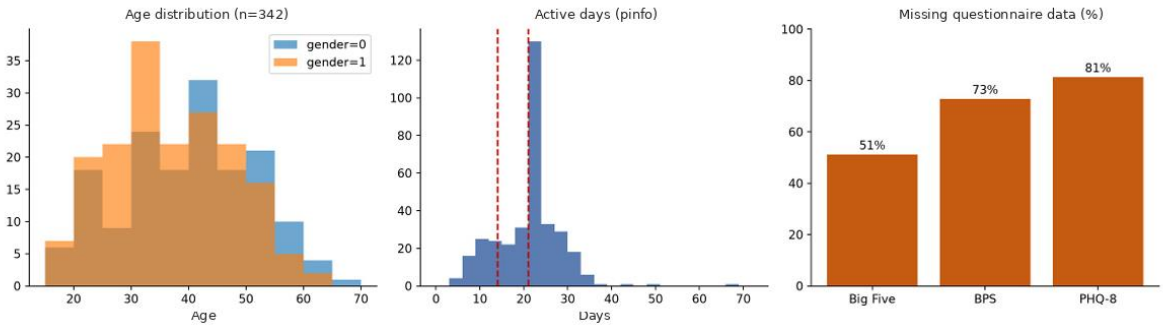


Figure 2. Participant age, active-day distribution and missing questionnaire data. Original figure; n = 342 for the participant profile.

3. Data Cleaning

Questionnaire fields are omitted from model features rather than imputed: Big Five is 51% missing, BPS 73% and PHQ-8 81%. Sensor-specific blanks are not automatically evidence of absent phone use.

Issue	Reported evidence	Treatment
Participant eligibility	10 users with fewer than 7 valid days; 2 users with about 0.2 unlocks/day and clock problems	Require screen events, ≥7 valid days, ≥7 active screen days and ≥1 mean daily unlock; retain 330 users.
Clock errors and duplicates	657 events in 6 users; extreme deviation about 13,000 days	Filter to participant start_date–last_active with ±2-day tolerance; remove exact duplicates.
Repeated Unlocked events	11.8% of unlock records; concentrated in 29 users	Count new Unlocked-to-first-Off sessions. Reject blanket 10-second deduplication: ≤10-second repeats are 1.9%, and two users lost 15% under that rule.
Overlong sessions	1,611 over 3 hours; 0.4% of sessions and 1.3% of valid windows touched	Cap feature contributions at 3 hours; exclude invalid sessions retrospectively from label/EDA screen duration.
Sensor versions	63 users with changed notifications only; 3 without On events	Use a common event schema; keep notifs optional.
Behavioural extremes	Up to 800 minutes/day and 227 unlocks/day	Retain these as observed behavioural variation rather than automatically deleting outliers.

Eligibility uses has_screen_events, valid_days, active_days_screen and mean_daily_unlocks. The two specifically identified low-activity users are ub91a17 and ueda26f. For repeated unlocks, the reported example ua297b9 has 204 raw daily unlocks versus 48 reconstructed sessions. The implementation preserves separate causal feature values and retrospective clean values, clean_screen_s is not a model input.

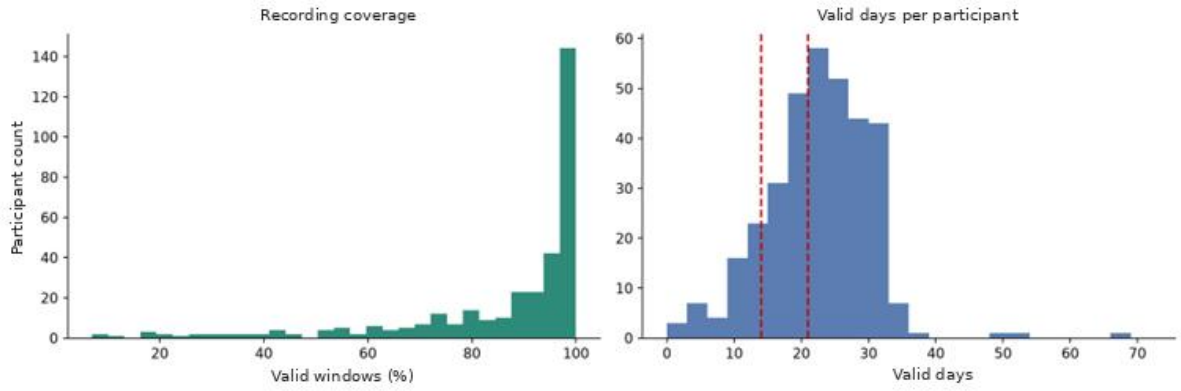


Figure 3. Recording coverage and valid-day counts in the prepared data. Original figure.

4. Exploratory Data Analysis

The data report summarizes 330 eligible participants, 366,747 valid sessions and 372,349 valid windows. A median participant has valid heartbeat coverage in 95% of windows. There are 295 participants with at least 14 valid days and 209 with at least 21 valid days.

The session-duration distribution has concentrations near 1 second and around 30 seconds, 1 minute and 300 seconds. The report associates the latter with screen-timeout settings, screen-on duration is therefore an imperfect measure of active engagement. Median daily screen duration is 139 minutes, with an extreme of 800; median daily unlock count is 41, ranging from 1 to 227 across users.

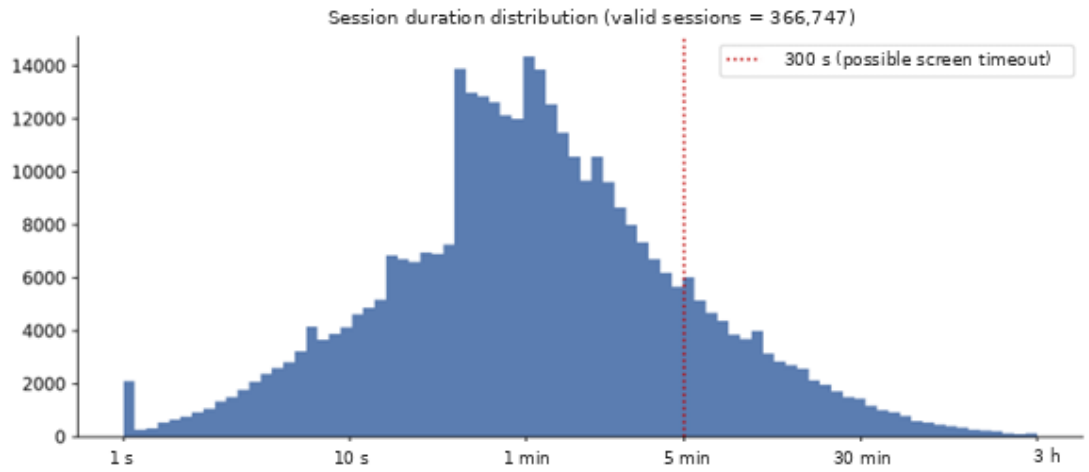


Figure 4. Duration distribution for 366,747 valid sessions. Original figure; logarithmic duration axis.

Dataset v1 defines three operational proxy components and their union, lab_any. In the script, the long-session label uses the total duration of a valid session intersecting the window (lbl_max_session_s), not only that session's overlap within the window. The within-window longest_run_s feature is a separate quantity.

Component	Default rule	Positive valid windows
Long session	Intersecting valid session ≥20 minutes	8.2%
Night use	00:00–05:00 local time and ≥5 minutes clean screen time	2.3%
Rapid unlock	≥6 session starts per 30-minute window	2.5%
Any component	Logical OR of the three labels	11.5%

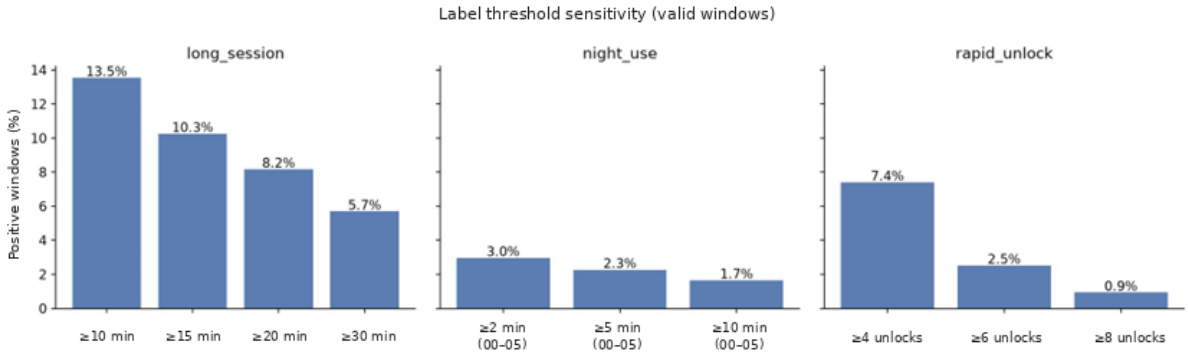


Figure 5. Proxy threshold sensitivity in the prepared data. Original values and graphics retained.

Long-session cutoffs of 10, 15, 20 and 30 minutes yield 13.5%, 10.3%, 8.2% and 5.7% positives. Night-use cutoffs of 2, 5 and 10 minutes yield 3.0%, 2.3% and 1.7%; rapid-unlock cutoffs of 4, 6 and 8 yield 7.4%, 2.5% and 0.9%. Changing the session cap from 2 to 6 hours changes long-session prevalence from 7.6% to 9.0% and night use from 2.1% to 2.5%. These results describe proxy-label sensitivity during data preparation.

Fixed thresholds affect users differently: some users have about 60% of windows labelled positive; 38 users are below 2%, while 41 are above 25%. This motivates testing personalization, but does not demonstrate that personalization already works.

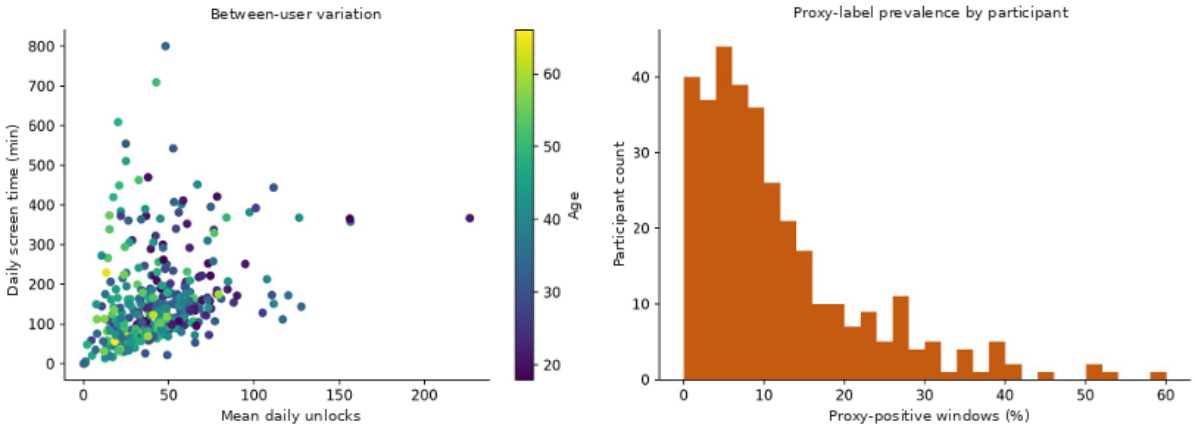


Figure 6. Between-user variation and participant-level proxy prevalence. Original figure.

5. Feature Engineering

Thirteen candidate fields are generated, including the optional notification count; removing notifs leaves 12 candidate fields.

Feature fields	Definition and rationale
screen_s	Causally capped screen duration in the current window, in seconds.
sessions_started	New session starts; repeated unlocks during an open session are not counted again.
longest_run_s	Longest continuous screen-use overlap within the window, at most 1,800 seconds.
app_switches	Application transitions, excluding launcher, keyboard and system prefixes (+, =, -).
prev_session_s	Duration of the latest session ended by the window start; the script includes an end exactly on that boundary.
hour, dow	Local hour 0–23 and weekday, Monday = 0.
is_night, is_weekend	Local night interval 00:00–05:00 and weekend flags.
dev_screen, dev_unlocks	Past-only rolling deviations from the participant's own usage distribution.
hist_days, notifs	Cumulative valid-window days and optional notification count.

Excluded from the model-input set are lab_long, lab_night, lab_rapid, lab_any, unlocks_raw, clean_screen_s and identity/metadata fields uuid, w_start_s, day_index, valid and target_ok. Target fields y_next_long, y_next_night, y_next_rapid and y_next_any contain the immediately following window's labels only when both windows are valid.

5. Feature Engineering - Validation

Data preparation validation includes a truncation test on 488 windows from 30 users, including windows intersected by overlong sessions. Removing all raw events after a tested window's end and recomputing features gave zero mismatches at 10^{-6} tolerance (488/488). Manual reconstruction of 15 windows from five users matched all session counters, with a maximum screen-duration rounding difference of 2.0 seconds.

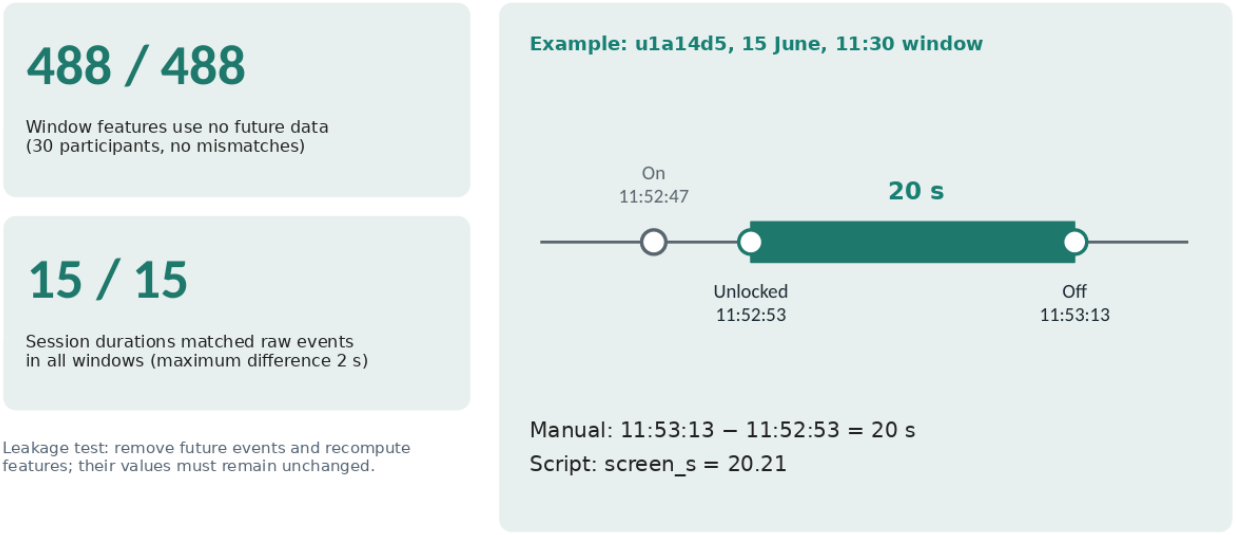


Figure 7. Reported feature-leakage and manual reconstruction checks. Original figure.

6. Data Transformation

The timeline is discretized into 1,800-second windows. Hour and weekday are integer-coded; night, weekend, proxy labels and targets use binary coding. For dev_screen and dev_unlocks, invalid windows are masked and values are shifted by one window before a 336-window rolling mean and standard deviation are computed. At least 48 valid observations are required within that seven-calendar-day window; they need not constitute a single complete day. Deviations divide by standard deviation plus 1e-6 and are clipped to [−10, +10]. hist_days is the cumulative valid-window count divided by 48.

```
# Excerpt from mpu_pipeline.py: past-only personal deviations
for col, out in [("screen_s", "dev_screen"), ("sessions_started", "dev_unlocks")]:
    x = W[col].where(W.valid)
    mu = x.shift(1).rolling(7 * DAY_WINDOWS, min_periods=DAY_WINDOWS).mean()
    sd = x.shift(1).rolling(7 * DAY_WINDOWS, min_periods=DAY_WINDOWS).std()
    W[out] = ((x - mu) / (sd + 1e-6)).clip(-10, 10)
W["hist_days"] = W.valid.cumsum() / DAY_WINDOWS

# Shift on the original window sequence; require an adjacent valid target.
W["target_ok"] = W.valid & W.valid.shift(-1, fill_value=False)
for lab, tgt in zip(LABEL_COLS, TARGET_COLS):
    W[tgt] = W[lab].shift(-1).where(W.target_ok)
```

Part 2 | Model Exploration

Preliminary model exploration evaluates a fixed rule, Logistic Regression and Random Forest using four numerical features from the research adapter. The participant coverage, preprocessing policies and operational target used for these experiments are specified below.

1. Model Selection

Random Forest is the main candidate selected for further investigation, compared against a transparent fixed rule and Logistic Regression. The task is to predict an operational elevated-use proxy in the next 30-minute window from completed-window behaviour. The present choice concerns offline research, not a final deployed or clinically validated model.

Random Forest can represent nonlinear relationships and feature interactions in the numerical window table. Its depth and leaf-size settings constrain complexity. It achieved the highest F1 and average precision among the current candidates. However, its F1 advantage over the simple rule is modest, and false-positive burden remains important. It is less straightforward to interpret than a rule or linear model. Explanations, model size and mobile latency require separate validation; statistical significance and production suitability have not been established.

Logistic Regression provides a lightweight linear reference with regularization. It helps establish whether nonlinear modelling adds value. Its limitation is the linear decision function in the current feature space. Its initial decision threshold produced comparatively high precision and low recall. Lowering the threshold recovered more proxy-positive windows but also increased false-positive predictions.

The fixed rule is transparent and inexpensive: a positive prediction is made when current screen-active time or current unlock count reaches the configured research cutoff. Its score is the fraction of rules met, NOT a probability. It cannot learn complex feature interactions, but its competitive F1 means that a benefit from ML must be demonstrated rather than assumed.

XGBoost is planned for subsequent exploration. It has NOT been trained or evaluated in these results. Personal-only and sample-weighted personalization experiments are also future work.

2. Model Training

Data preparation and feature engineering

The local Mobile Phone Use Dataset contains 342 participant CSVs, 31,562,291 event rows and a shared 80-column raw schema. Each row is a sensor event, not a training example. Empty fields for unrelated sensors are structural blanks, the complete raw row is not fed directly into a classifier. Events are sorted per user by ts_raw, a Unix timestamp in milliseconds.

The research adapter reconstructs Unlocked-to-Off sessions. On alone neither starts nor ends an unlocked-use session. Foreground-app transition counting excludes launcher (+), keyboard (=), system (-) and empty observations. Session intervals are intersected with fixed, half-open 30-minute UTC windows.

Four model features are used:

- *screen_active_seconds:* unlocked-to-Off activity intersecting the completed window;
- *unlock_count:* observed unlock events within the completed window;
- *app_switch_count:* transitions among retained foreground-app observations;
- *previous_session_duration_seconds:* duration of a reliable session ending strictly before the prediction boundary.

Research quality policies exclude affected observations for recording gaps longer than 30 minutes, incomplete/ambiguous sessions and sessions exceeding 120 minutes. After a recording gap, screen state must be observed again; unknown state is not silently treated as non-use. The configured 2016 event-year filter excludes 642 other-year events. These policies are explicit assumptions requiring sensitivity checks, not validated live Android coverage.

The full dataset yields 444,012 candidate windows, 334,165 valid windows and 319,614 valid adjacent-window pairs. There are 109,847 invalid windows; another 14,551 valid windows lack a valid immediately following target window or are terminal windows. No user is fully excluded. Targets are shifted on the full grid *before* invalid windows are removed, so a later available window cannot replace the immediate next window.

The proxy is positive when the *next* window's longest continuous unlocked-use portion OR unlock count reaches its corresponding fitting-data 90th-percentile cutoff. Floors of one second and one unlock prevent zero use from automatically becoming positive when a percentile is zero. These are operational cutoffs, not validated unwanted-use thresholds. Night use is omitted while local-time reconciliation remains unresolved. Participant IDs, future-window measurements, questionnaire scores and whole-study ESM summaries are not model features.

Preprocessing and hyperparameters

Logistic Regression uses a Pipeline with median imputation, missingness indicators, StandardScaler and the classifier. All learned transformations are fitted on fitting users only. Entirely missing training columns are retained with a zero numerical placeholder and an indicator; this is not an assertion that observed usage was zero. Validation/test statistics never fit the transformations.

Logistic Regression: *C=1.0, solver=lbfgs, max_iter=1000, random_state=42, class_weight=None. The initial wrapper decision threshold is 0.50. Random Forest: n_estimators=100, max_depth=8, min_samples_leaf=5, min_samples_split=2, max_features=sqrt, criterion=gini, bootstrap=True, class_weight=balanced, random_state=42, n_jobs=1.*

It uses training-only median imputation with indicators and no scaling. Class weights are derived during fitting. Fixed rule: any one of the two current-window rules is sufficient. Its cutoffs derive from the fitting population's proxy definition, never from test outcomes.

No classifier hyperparameter search was performed. These are documented starting configurations, not optimally tuned settings.

Recorded environment: *Python 3.11.15, NumPy 2.4.6, pandas 3.1.0, scikit-learn 1.9.1, matplotlib 3.11.2*.

Cross-validation and decision selection

The initial full-data comparison used three-fold ***GroupKFold*** by participant: 229/113, 227/115 and 228/114 training/test users. Each user was tested once, with no user shared between training and test in a fold. Proxy cutoffs were learned only from that fold's training users.

A separate decision-threshold study kept the same outer folds. Within each development group, ***GroupShuffleSplit*** reserved 20% of USERS for validation (random_state=42). This is not necessarily 20% of rows. Both proxy cutoffs and model fitting used only fitting users; validation/test labels used those frozen proxy cutoffs. The partitions were:

Fold 1: 183 fitting / 46 validation / 113 test users; 173040 / 40049 / 106525 rows.

Fold 2: 181 fitting / 46 validation / 115 test users; 170854 / 42200 / 106560 rows.

Fold 3: 182 fitting / 46 validation / 114 test users; 167094 / 45991 / 106529 rows.

For Logistic Regression and Random Forest, the decision threshold maximized validation F1 over the predefined grid 0.05, 0.10, ..., 0.95. Exact ties prefer the higher threshold. Outer-test labels do not select the threshold. Models are not refitted after selection: the default-0.50 control and selected policy therefore use identical test scores and labels. Positive decisions in this study use score >= threshold. The fixed rule is not tuned.

Validation-selected decision thresholds:

Fold 1: Logistic Regression 0.20; Random Forest 0.65.

Fold 2: Logistic Regression 0.15; Random Forest 0.55.

Fold 3: Logistic Regression 0.20; Random Forest 0.60.

Because the fitting population is smaller in this second study, its proxy cutoffs also differ from the initial comparison. Decision-selection effects must be compared with the 0.50 control WITHIN this study. Changes between the two experiments cannot be attributed solely to a decision threshold.

3. Model Evaluation

Metrics are precision, recall, F1, average precision (AP, used here as PR-AUC), ROC-AUC, confusion matrices and false-positive predictions per evaluated participant-UTC-day. AP is not trapezoidal PR integration. Tables are unweighted means of outer-test fold metrics. Figures pool out-of-fold predictions; proxy cutoffs may differ by fold. Four users in the initial full-data experiment had only negative proxy labels; undefined per-user AP/ROC-AUC were left null rather than silently excluding users.

FP/day uses unique evaluated participant/UTC-date pairs, including partial days. It measures proxy false-positive prediction burden, NOT delivered Android notifications. No notification cap or quiet-hours simulation was applied.

Initial full-data comparison: 342 users and 319,614 window pairs. Unweighted outer-fold means.

Model	Precision	Recall	F1	AP	ROC AUC	FP/day
Rule	.510	.540	.525	.383	.706	3.811
Logistic Regression	.636	.319	.425	.529	.792	1.337
Random Forest	.425	.727	.536	.547	.796	7.244

Controlled decision-threshold study: all 342 outer-test users. Unweighted outer-fold means.

Model and policy	Prec.	Recall	F1	AP	ROC AUC	FP/day
LR default 0.50	.634	.317	.422	.527	.791	1.332
LR validation F1	.446	.677	.537	.527	.791	6.193
RF default 0.50	.418	.731	.532	.544	.796	7.458
RF validation F1	.477	.640	.544	.544	.796	5.202
Rule fixed	.509	.538	.523	.381	.706	3.803

Within the controlled study, Random Forest F1 increased from 0.532 to 0.544 while FP/day decreased from 7.458 to 5.202. Logistic Regression F1 increased from 0.422 to 0.537, but FP/day rose from 1.332 to 6.193. Maximizing F1 does not enforce an acceptable notification budget. RF remains the main research candidate; the advantage is modest and this is not a final deployment decision. An independent probability-calibration study has not been performed.

Selected-policy pooled confusion matrices use rows for observed 0/1 and columns for predicted 0/1. Exact counts are shown below.

Rule: [[220831, 33812], [29985, 34986]]

Logistic Regression: [[199492, 55151], [20968, 44003]]

Random Forest: [[208250, 46393], [23379, 41592]]

The selected-policy figures show the fixed rule and validation-selected LR/RF decisions. AP and ROC-AUC do not change when only the decision threshold changes on a fixed fitted model, because they use the scores. The selected-policy matrices differ from the default-0.50 control results.

Limitations: The target is behavioural proxy usage, not reported unwanted use or addiction. Historical data may not transfer to current users in Türkiye. Proxy/quality sensitivity, personalization, user-feedback agreement and mobile inference remain untested. The dataset has already been explored; these folds are not an untouched external deployment cohort. **Next work includes sensitivity checks, a user-appropriate false-alert objective, XGBoost and the planned personalization comparisons.**

Model Evaluation figures

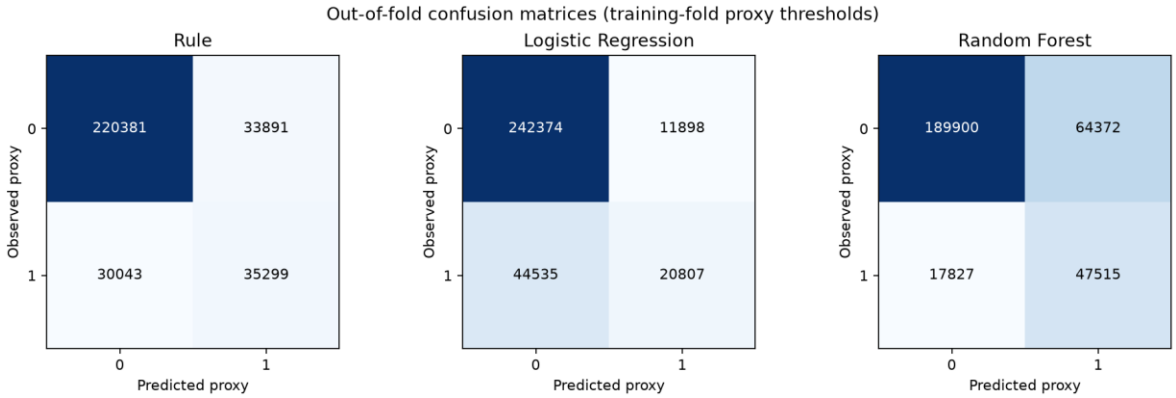


Figure 8. Initial full-data comparison, pooled out-of-fold confusion matrices. Original repository figure.

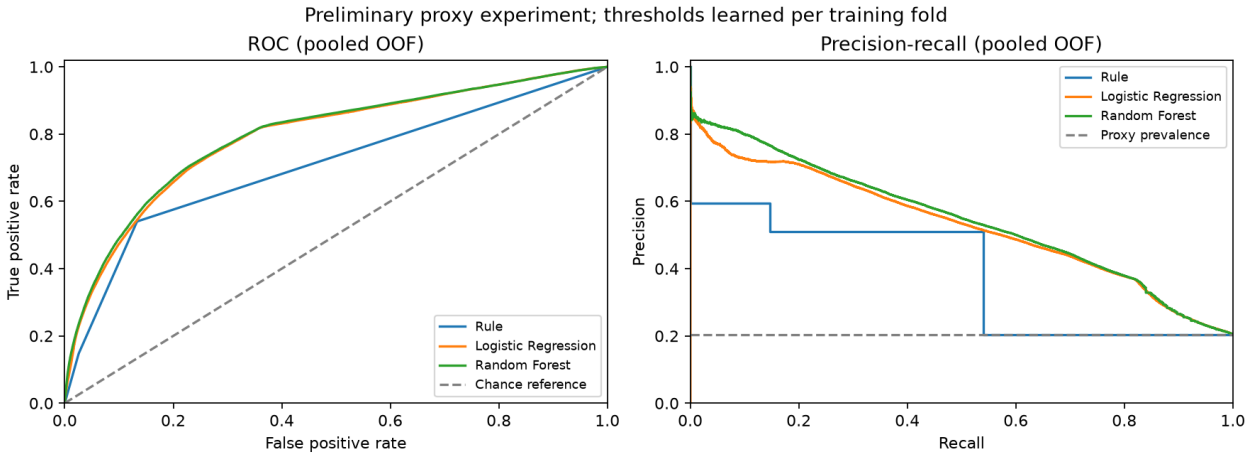


Figure 9. Initial full-data comparison, pooled out-of-fold ROC and precision–recall curves. Original repository figure.

The metric tables report unweighted means across folds. Pooled figures summarize all out-of-fold predictions, so their displayed values need not equal the table means. Proxy cutoffs can differ by fold.

Model Evaluation selected policies

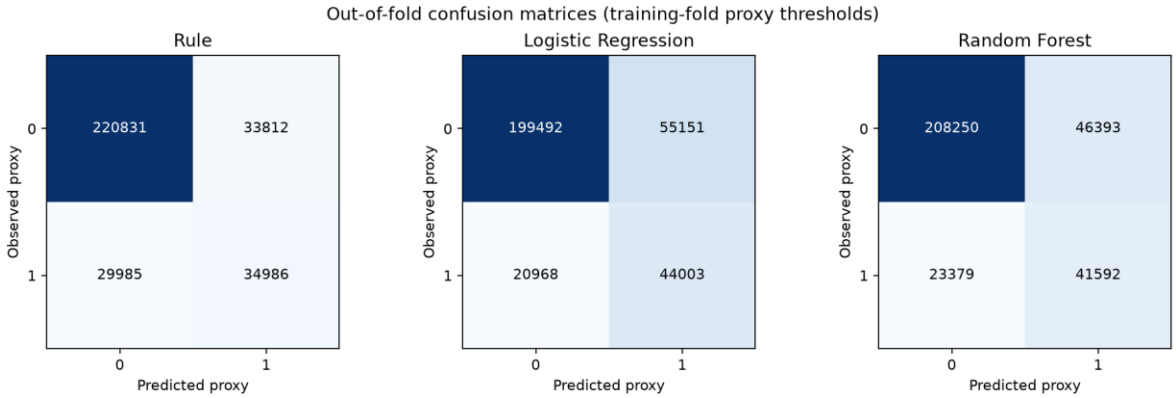


Figure 10. Controlled study: fixed rule and validation-selected LR/RF decisions. These are not the default-0.50 control matrices.

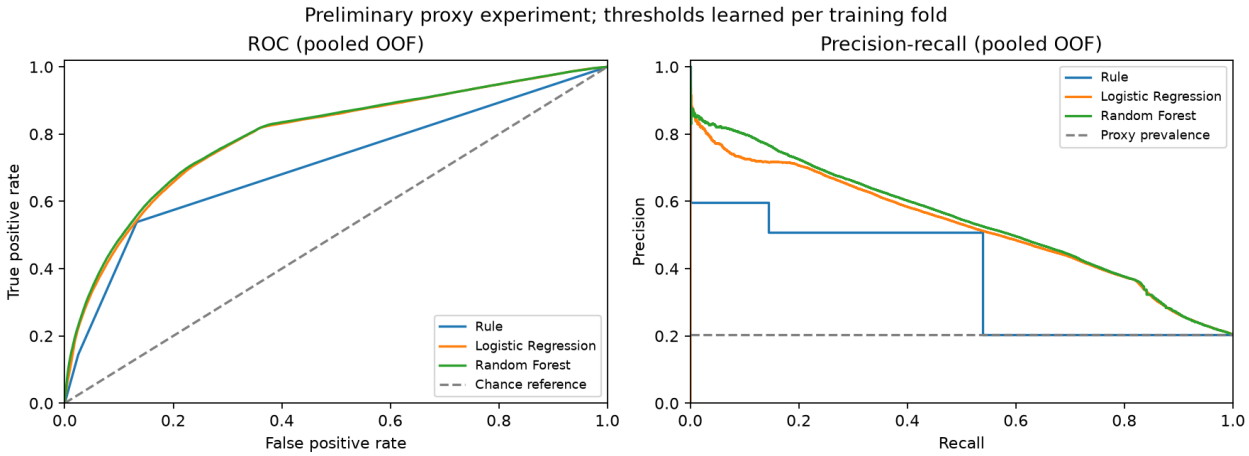


Figure 11. Controlled study: pooled out-of-fold score curves. Changing only the decision threshold does not change AP or ROC-AUC for the same fitted model.

Random Forest remains a research candidate. Its selected-policy F1 of 0.544 versus 0.523 for the fixed rule is a modest difference; false-positive burden, calibration, sensitivity and mobile inference still require investigation.

4. Code Implementation

Part 1 includes the relevant data preparation code. The model code below comes from the supplied research draft and the corresponding repository modules. These code blocks show parts of functions and cannot run on their own. The required imports and other variables are defined in the complete modules.

Window intersection and adjacent targets

```
# src/models/research_data.py
# Intersect reconstructed sessions with each half-open window.
for start, end in sessions:
    for i in indices(start, end):
        overlap = max(0, min(end, int(ends[i])) - max(start, int(starts[i]))) / 1000
        active[i] += overlap
        longest[i] = max(longest[i], overlap)

# Shift BEFORE removing invalid windows: never skip a gap to find a target.
frame["next_screen_active_seconds"] = frame["screen_active_seconds"].shift(-1)
frame["next_unlock_count"] = frame["unlock_count"].shift(-1)
frame["next_longest_session_seconds"] = pd.Series(longest).shift(-1)
paired = valid & np.r_[valid[1:], False]
result = frame.loc[paired].reset_index(drop=True)
```

Fitting user proxy cutoffs

```
# src/models/real_data_exploration.py
# training contains fitting users only; floors prevent zero-use positives.
def fit_proxy_thresholds(training: pd.DataFrame, quantile: float) -> dict:
    if not 0 < quantile < 1:
        raise ValueError("proxy quantile must be between 0 and 1")
    return {
        "longest_session_seconds": max(1.0, float(
            training["next_longest_session_seconds"].quantile(quantile))),
        "unlock_count": max(1.0, float(
            training["next_unlock_count"].quantile(quantile))),
    }
```

Training only transformations

```
# src/models/baselines.py, inside build_logistic_pipeline
# Imputer and scaler are fitted with the classifier on fitting users only.
return Pipeline([
    ("imputer", SimpleImputer(strategy="median", keep_empty_features=True,
                              add_indicator=True)),
    ("scaler", StandardScaler()),
    ("classifier", LogisticRegression(C=C, max_iter=max_iter,
                                      random_state=random_state)),
])
```

Validation only decision selection

```
# src/models/threshold_exploration.py, inside select_decision_threshold
# Validation F1 chooses the decision threshold; exact ties favor the higher one.
rows = [{"threshold": float(t),
         "validation_f1": float(f1_score(y, scores >= t, zero_division=0))}
        for t in grid]
best = max(rows, key=lambda row: (row["validation_f1"], row["threshold"]))
return best["threshold"], rows
```

participant_splits uses outer GroupKFold and inner GroupShuffleSplit and checks disjoint user sets. compare_thresholds fits models, selects thresholds using validation predictions, then evaluates frozen models on outer-test users. Split and candidate-grid records preserve this ordering.

Reproduction and validation evidence

```
.venv/bin/python -m src.models.dataset_audit
.venv/bin/python -m src.models.real_data_exploration --participants 342 --folds 3 --output
reports/model_exploration_all_342 --workdir data/processed/model_exploration_all_342
.venv/bin/python -m src.models.threshold_exploration
.venv/bin/python -m pytest -q --tb=short
```

Run these commands from the repository root with the local dataset present. The recorded model work reports 63 passed automated tests and checks of source hashes, three-way participant isolation, validation-selected thresholds, identical control/selected scores, confusion matrices and test metrics. Raw logs, row-level predictions and split manifests remain under ignored local paths. Earlier reports were preserved. These results do not establish addiction reduction or successful Android deployment.

Next steps

The next stage will establish a versioned common input schema and evaluation protocol, then repeat model comparisons on the agreed feature table. Feature definitions, participant eligibility, timestamps, quality rules and proxy labels will be recorded with each experiment.

The planned general, sample-weighted personalized and personal-only comparisons at 1, 3, 7 and 14 days of history remain future work. XGBoost is also unevaluated. Current evidence supports further model exploration, not completion of the project's personalization research question.

Sources and implementation records

Data preparation: İsmail Hakkı Sarıoğlan, Grup8_Veri_Hazirlama_ve_Model_Kesfi_Raporu.docx, Part 1; accompanying mpu_pipeline.py. Data figures retain the original plotted content; labels are translated into English.

Model development: Hayrettin Kaan Özsoy, Model Exploration draft, evidence snapshot

Dataset reference: CRAWDAD telefonica/mobilephoneuse, DOI 10.15783/jh3s-h006.

Repository: https://github.com/edasaruhan/SIC_AI_18_Capstone_Group_8

Model results: reports/model_exploration_all_342/results.json and reports/model_exploration_thresholds/results.json; corresponding confusion_matrices.png and roc_pr_curves.png.

Implementation: src/models/research_data.py, real_data_exploration.py, baselines.py and threshold_exploration.py. Current research status: documents/model_development_plan.md.

Appendix | Data coverage and manual checks

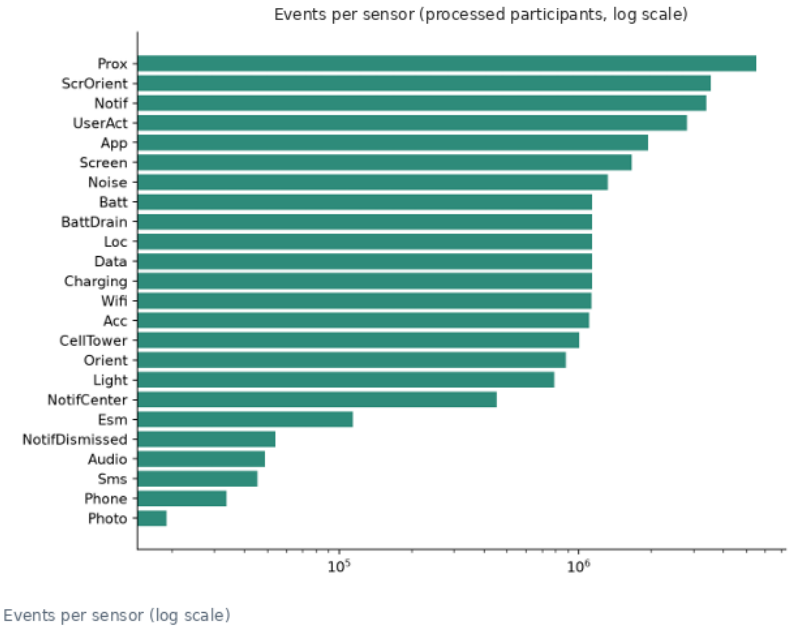


Figure 12. Sensor-event coverage in the prepared data. Original figure; logarithmic event-count axis.

Manual reconstruction checks

Original reported values from elle_kontrol_tablosu.csv. Counter agreement was reported in all 15 windows, screen-duration differences reflect the rounded manual values.

User	Window	Time	Manual s	Script s	Diff s	Starts	Status
u072142	145	18:00	1671.0	1671.58	0.6	4 / 4	Matched
u072142	260	03:30	174.0	173.97	-0.0	1 / 1	Matched
u072142	682	22:30	446.0	446.07	0.1	0 / 0	Matched
u0cba7d	373	13:30	418.0	418.17	0.2	12 / 12	Matched
u0cba7d	863	18:30	255.0	254.69	-0.3	4 / 4	Matched
u0cba7d	999	14:30	668.0	667.06	-0.9	8 / 8	Matched
u11abb4	192	18:00	1741.0	1741.11	0.1	2 / 2	Matched
u11abb4	297	22:30	1770.0	1770.49	0.5	1 / 1	Matched
u11abb4	1063	21:30	355.0	355.36	0.4	2 / 2	Matched
u11cf22	863	13:00	342.0	342.35	0.4	8 / 8	Matched
u11cf22	1019	19:00	27.0	26.92	-0.1	2 / 2	Matched
u11cf22	1166	20:30	294.0	295.96	2.0	7 / 7	Matched
u1a14d5	311	11:30	20.0	20.21	0.2	1 / 1	Matched
u1a14d5	367	15:30	921.0	920.30	-0.7	2 / 2	Matched
u1a14d5	851	17:30	307.0	306.51	-0.5	1 / 1	Matched